

Claim frequency and burning cost across portfolio segments

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1. Data

`dataCar` contains 67,856 one-year vehicle insurance policies taken out in 2004 or 2005, including exposure e_i (exposure), claim counts n_i (numclaims) and claim amounts (claimcst0) X_i .

All portfolio calculations are performed in SQL.

2. Goal

The aim is to investigate how claim frequency, claim severity, and burning cost differ across portfolio segments, focusing specifically on driver age, geography, and vehicle value. For each segment, we calculate

$$\text{frequency} = \frac{\sum n_i}{\sum e_i}, \quad \text{severity} = \frac{\sum X_i}{\sum n_i}, \quad \text{burning cost} = \frac{\sum X_i}{\sum e_i}.$$

3. Setup

First, we grab the packages and the data set `dataCar` from `insuranceData`:

```
library(insuranceData); library(duckdb); library(DBI); library(knitr)

data(dataCar)
```

Then, we start a database DB and copy `dataCar` into it. We also define the shortcut `SC` that sends SQL to DB and gives back the result as a data frame (it enables us to write SQL queries in R):

```
DB <- dbConnect(duckdb())
dbWriteTable(DB, "policies", dataCar, overwrite = TRUE)

SC <- function(sql) dbGetQuery(DB, sql)
```

4. Portfolio overview

Below is an overview of the portfolio. It contains 31,800.8 exposure years and 4,937 claims corresponding to a claim frequency of 0.1552 claims per exposure-year and a burning cost of 292.90 per exposure-year.

```
SC("SELECT
  COUNT(*) AS policies,
  SUM(exposure) AS exposure_years,
  SUM(numclaims) AS claims,
  sum(claimcst0) AS total_cost,
```

```
SUM(numclaims)/SUM(exposure) AS frequency,
SUM(claimcst0)/SUM(numclaims) AS severity,
SUM(claimcst0)/SUM(exposure) AS burning_cost
FROM policies;")
```

```
## policies exposure_years claims total_cost frequency severity burning_cost
## 1 67856 31800.82 4937 9314604 0.1552476 1886.693 292.9045
```

5. Driver age

How does claims cost vary across driver age groups? Looking at the table below, we see that the burning cost falls from 500.47 in the youngest age category (`agecat = 1`) to 205.26 in the oldest age category (`agecat = 6`). The lowest burning cost is observed in category 5 at 205.26.

The difference is driven by both components: group 1 has both higher claim frequency and severity.

```
SC("SELECT agecat,
SUM(exposure) AS exposure,
SUM(numclaims) AS claims,
SUM(numclaims)/SUM(exposure) AS frequency,
SUM(claimcst0)/SUM(numclaims) AS severity,
SUM(claimcst0)/SUM(exposure) AS burning_cost
FROM policies
GROUP BY agecat
ORDER BY agecat;")
```

```
## agecat exposure claims frequency severity burning_cost
## 1 1 2612.274 525 0.2009743 2490.234 500.4732
## 2 2 5891.871 1000 0.1697254 1984.841 336.8778
## 3 3 7409.457 1189 0.1604706 1793.194 287.7549
## 4 4 7616.542 1185 0.1555824 1810.382 281.6636
## 5 5 5171.009 648 0.1253140 1637.982 205.2621
## 6 6 3099.666 390 0.1258200 1752.740 220.5297
```

6. Geography

Do claims costs differ substantially between geographical areas? Looking at the table below, we see that Area F stands out with a burning cost of 461.96, which is substantially above the other areas.

Unlike the age effect, this difference is primarily driven by severity: Area F has a frequency only moderately above the portfolio average, but its average claim is substantially more expensive.

```
SC("SELECT
area,
SUM(exposure) AS exposure,
SUM(numclaims) AS claims,
SUM(numclaims)/SUM(exposure) AS frequency,
sum(claimcst0)/SUM(numclaims) AS severity,
SUM(claimcst0)/SUM(exposure) AS burning_cost
FROM policies
GROUP BY area
ORDER BY burning_cost DESC;")
```

	area	exposure	claims	frequency	severity	burning_cost
## 1	F	1735.992	305	0.1756921	2629.362	461.9581
## 2	E	2771.866	413	0.1489971	2103.687	313.4434
## 3	C	9578.494	1493	0.1558700	1919.429	299.1814
## 4	B	6297.848	1021	0.1621189	1758.369	285.0649
## 5	A	7597.101	1181	0.1554540	1754.247	272.7048
## 6	D	3819.518	524	0.1371901	1738.661	238.5270

7. Vehicle value

How does vehicle value relate to claims cost? Looking at the table below, we see that burning cost increases consistently with vehicle value, from 267.82 to 331.53.

The increase is driven by claim frequency rather than severity. More valuable vehicles in this portfolio generate claims more frequently, while their average claim costs remain relatively stable.

```
SC("SELECT
CASE
WHEN veh_value < 1 THEN '0-1'
WHEN veh_value < 2 THEN '1-2'
WHEN veh_value < 3 THEN '2-3'
ELSE '3+'
END AS value_band,
SUM(exposure) as exposure,
SUM(numclaims) AS claims,
SUM(numclaims)/SUM(exposure) AS frequency,
SUM(claimcst0)/SUM(numclaims) AS severity,
SUM(claimcst0)/SUM(exposure) AS burning_cost
FROM policies
GROUP BY value_band
ORDER BY value_band;")
```

	value_band	exposure	claims	frequency	severity	burning_cost
## 1	0-1	7637.979	1022	0.1338050	2001.566	267.8196
## 2	1-2	14990.305	2324	0.1550335	1863.119	288.8460
## 3	2-3	5255.398	894	0.1701108	1834.957	312.1460
## 4	3+	3917.136	697	0.1779361	1863.218	331.5338

8. Conclusion

Three distinct patterns emerge from the analysis:

- Driver age: Burning cost generally decreases with age, driven by both lower frequency and lower severity.
- Geography: Area F has particularly high burning cost, driven primarily by claim severity.
- Vehicle value: Higher vehicle values are associated with higher burning cost, driven primarily by claim frequency.

This illustrates analysis benefits from separating frequency and severity rather than looking only at total claims cost. Different segments can have similar overall risk for fundamentally different reasons.

The analysis is descriptive and does not establish causal relationships between the characteristics and claims costs. The variables may also interact with one another, so further modelling would be required to isolate their individual effects.